

Unsupervised Discovery of Market Regimes Using Static and Temporal Models

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Date: December 29, 2025

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Motivation

Financial markets exhibit distinct periods of behavior characterized by differing levels of risk, volatility, and macroeconomic conditions. These periods - usually referred to as market regimes - play a crucial role in portfolio construction, risk management, and macroeconomic analysis in general. Accurately identifying market regimes is a fundamental issue in economics, finance and applied data science.

Traditional approaches to identify and characterise regimes often rely on predefined economic rules or labels, such as recession indicators, volatility thresholds, etc. While valuable and interpretable, these methods set strong assumptions about regimes and often fail to capture gradual transitions or latent market dynamics. On the other hand, modern machine learning approaches are increasingly applied to financial data, but frequently focus on predictive performance rather than interpretability or stability, limiting their usefulness in understanding underlying market behavior.

Unsupervised learning offers a good alternative by allowing regimes to emerge directly from data without predefined labels. However, the extent to which different unsupervised modeling assumptions affect the stability, interpretability, and economic relevance of discovered regimes remains insufficiently explored in applied settings. In particular, systematic comparisons between static clustering methods (Gaussian Mixture Models, GMM), temporally-aware probabilistic models (Hidden Markov Models, HMM), and temporally constrained hierarchical clustering (TCHC) on macro-financial indicators are relatively scarce in applied settings.

This project addresses this gap by comparing different unsupervised regime discovery methods using weekly macro-financial data between 1995 and 2024. By evaluating *Gaussian mixture models*, *hidden Markov models*, and *time-aware clustering techniques*, the study focuses on *regime stability*,

cross-model agreement, and economic interpretability rather than predictive accuracy.

The goal is to provide a clearer understanding of how modeling choices influence the regimes that emerge and their potential usefulness for economic and financial analysis. In simple words, I want to figure out how different unsupervised modeling decisions change the way markets are partitioned into regimes, and whether those regimes are stable, interpretable, and meaningful for analyzing market risk and behavior.

Research questions

This study aims to address the following research questions:

- How stable are the market regimes identified by different unsupervised learning methods when applied to weekly macro-financial data?
- Does incorporating explicit temporal structure (e.g., Hidden Markov Models) lead to more persistent and interpretable regimes compared to static clustering methods (e.g., Gaussian Mixture Models)?
- To what extent do different unsupervised models agree on the timing and structure of identified market regimes?
- Are the identified regimes associated with distinct economic characteristics, such as differences in volatility, drawdowns, and market stress indicators?
- How sensitive are the identified regimes to modeling choices such as the number of regimes and feature selection?

Scope & Non-Goals

Scope

This study focuses on the unsupervised identification and evaluation of market regimes using weekly macro-financial indicators. The analysis is conducted at the aggregate market level, with the S&P 500 serving as a proxy for overall equity market behavior. The time horizon spans multiple market cycles in order to capture a range of economic conditions, including periods of expansion, stress, and crisis.

The modeling scope is limited to unsupervised learning methods commonly used for regime discovery, including static clustering approaches (e.g., Gaussian Mixture Models) and temporally-aware probabilistic models (e.g., Hidden Markov Models). A single additional time-aware clustering method is included to provide methodological contrast Temporally Constrained Hierarchical Clustering. All models are evaluated under a consistent experimental framework, using the same data, features, and evaluation criteria.

Evaluation emphasizes regime stability, cross-model agreement, and economic interpretability, rather than predictive accuracy. Regimes are assessed based on their persistence over time and their association with economically

meaningful characteristics such as volatility, drawdowns, and market stress indicators.

Non-goals

This project explicitly does not aim to:

- Predict future asset prices or returns
- Develop or evaluate trading strategies
- Generate alpha or optimize investment performance
- Perform supervised learning using predefined regime labels
- Apply deep learning or high-frequency modeling techniques
- Analyze individual securities or firm-level data

These exclusions are intentional and reflect the project's focus on understanding and evaluating unsupervised regime discovery methods, rather than forecasting or decision-making applications.

Data Description

Data Sources

The research uses publicly available macro-financial data sets that ensures transparency and reproducibility. Equity market behavior is represented using the S&P 500 index, while market uncertainty and stress are captured through the CBOE Volatility Index (VIX). Macroeconomic conditions are incorporated via interest rate term structure measures sourced from the Federal Reserve Economic Data (FRED) database.

Specifically, the dataset includes:

- S&P 500 Index (price levels and trading volume), sourced from Yahoo Finance
 - CBOE Volatility Index (VIX), sourced from Yahoo Finance
 - 10-year minus 2-year U.S. Treasury Yield Spread, sourced from FRED
- Optional credit market indicators (e.g., BAA–AAA corporate bond spreads) are considered when data availability and quality permit.

Time Period & Frequency

All series are converted to a weekly frequency, spanning the period 1995–2024, subject to data availability.

Modelling approach

The project applies three complementary unsupervised methods to identify market regimes:

- Gaussian Mixture Model (GMM) – a static clustering method*
- Hidden Markov Model (HMM) – a temporally-aware probabilistic model*

- Temporally Constrained Hierarchical Clustering (TCHC) – a soft time-aware clustering method

These methods are selected to span a spectrum of temporal assumptions:

- GMM assumes each observation is independent
- HMM explicitly models regime persistence and transition probabilities
- TCHC encourages contiguous regime assignments without full probabilistic modeling

All models are applied to the same weekly macro-financial dataset to ensure comparability.

**Utilizing Full Covariance matrices to capture feature correlations. e.g., VIX vs. Yields*

Evaluation strategy

Because the project focuses on unsupervised regime discovery, evaluation emphasizes stability, interpretability, and robustness, rather than predictive accuracy. The key idea: A good regime model identifies persistent, economically meaningful market states that are consistent across methods and parameter choices. Evaluation is therefore divided into four pillars:

- Regime Stability & Quality: Measures whether the identified regimes are persistent and meaningful over time using mean regime duration (average number of consecutive weeks per regime), transition frequency (how many times do regime switch), entropy of regime transitions (low entropy signaling more predictable regimes, high entropy signaling otherwise). Our goal is obviously high stability that would reflect true market structure, rather than useless noise.
- Cross-Model Agreement: Assesses whether different models produce consistent regime structures using ARI (Adjusted Rand Index) and visual alignment of regime sequences (timeline plots). Our aim is high agreement across models that would increase confidence that found regimes are robust and are not simply model artifacts.
- Economic Interpretability: Tests whether regimes correspond to economically meaningful patterns using average and standard deviation of returns per regime, volatility (realized and VIX) per regime, MDD (Maximum drawdowns per regime), correlation with macro stress indicators (e.g., yield spreads). Our goal is simply being able to explain model findings and differentiate between the periods of high and low market stress (i.e. we should have an intuitive financial explanation).
- Robustness Check: basically evaluates sensitivity to the model choices using number of regimes ($k = 2 - 6$), feature selection variations, temporal window adjustments (for TCHC). Our results should not change under small model modifications. Besides, to address look-ahead bias, we will perform an expanding window analysis (e.g., fit on 1995–2010, predict 2011)

to test if regimes identified in-sample generalize to out-of-sample data. Expanding window analysis is used here to test whether regimes identified in-sample generalize to out-of-sample periods, focusing on stability rather than predictive returns.

Summary

Evaluation focuses on:

<i>Pillar</i>	<i>Why does it matter?</i>	<i>How do we measure it?</i>
Stability	Are regimes persistent?	Duration, Transitions, Entropy
Cross-Model Agreement	Are results robust across methods?	ARI, Visual Timelines
Economic Interpretability	Are regimes meaningful?	Volatility, Drawdowns, Stress Correlation
Robustness	Are findings sensitive to choices?	K variation, Features, TCHC parameters, Expanding Window.

**No predictive metrics (accuracy, R^2 , Sharpe) are used, consistent with the unsupervised focus.*

Definition of success

The project will be considered successful if identified regimes are stable, interpretable, and consistent across models, and the findings are reproducible and robust to variations in model parameters and feature selection.